External-Space Self-Consistency Law

A speculative theoretical paper on a hypothetical temporal quantum machine

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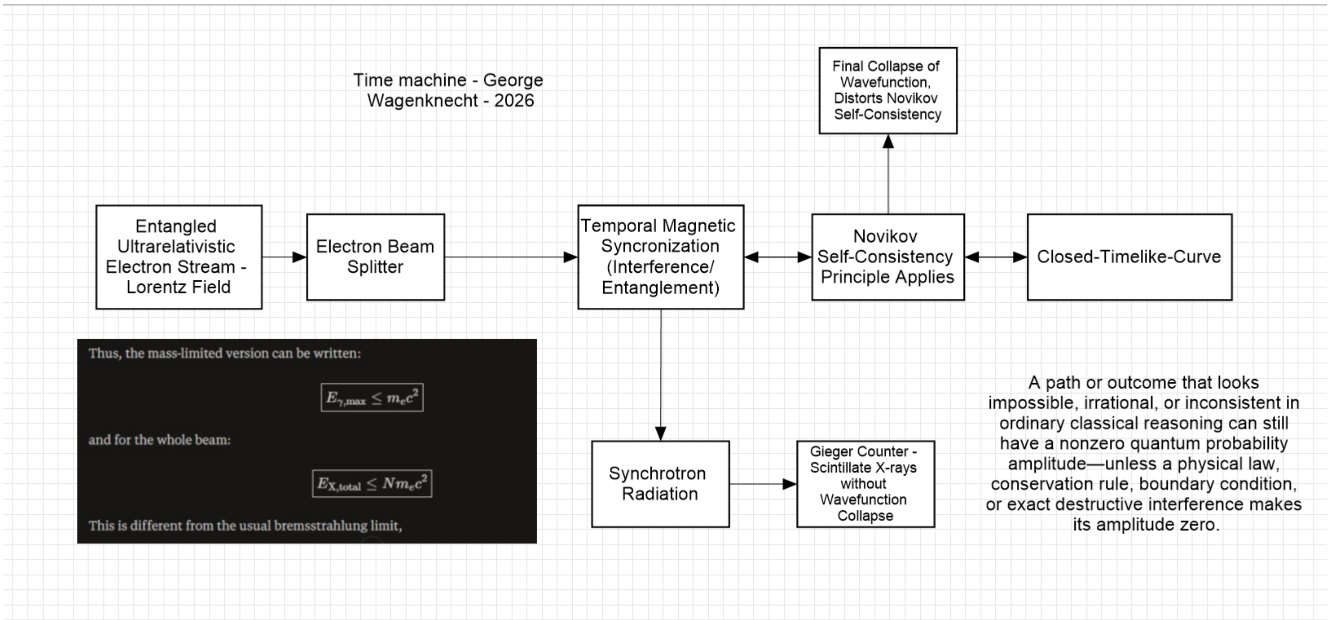

Figure 1. Conceptual architecture of the proposed machine, combining an electron-beam stage, temporal synchronization, closed-timelike-curve self-consistency, radiation, and measurement.

Abstract

This paper proposes a speculative principle called the *External-Space Self-Consistency Law*. The principle extends the intuition of self-consistency in closed-timelike-curve models into a description in which the machine does not literally rewrite an already established past. Instead, possible configurations of an external physical space are represented by amplitudes, and a global consistency condition determines which configurations can retain nonzero amplitude. The central proposal is that an externally realized configuration must remain compatible with the complete spacetime constraints imposed by the machine. Configurations that cannot close consistently are suppressed or excluded rather than retroactively replacing an observed past. The paper develops a mathematical toy model, relates it to established CTC literature, and distinguishes the proposal from experimentally demonstrated physics. It is intended as a conceptual research framework, not as a claim that macroscopic time machines or closed timelike curves have been experimentally realized.

1. The Proposed Machine

The machine shown in Figure 1 is treated here as a conceptual device rather than an experimentally validated time machine. Its proposed chain consists of an entangled or temporally correlated electron stream, an electron-beam splitter, a temporal magnetic synchronization stage, a closed-timelike-curve constraint, and a final measurement stage. Synchrotron radiation and X-ray detection are included as possible observable channels. The important theoretical element is not the individual hardware block, but the hypothesized global constraint connecting the machine's input, interaction region, and external configuration.

2. External Space as an Amplitude Space

Let the possible externally observable configurations be denoted by |E■■. A quantum description can be written schematically as

$$|\Psi\blacksquare = \Sigma\blacksquare\ A\blacksquare\ |E\blacksquare\blacksquare$$

where A■ is the complex amplitude associated with external configuration E■. The corresponding probability, when the states form an appropriate measurement basis, is proportional to |A■|². The proposed law does not assert that the past configuration is subsequently edited. Instead, it places a constraint on which complete configurations can have nonzero amplitude.

3. External-Space Self-Consistency Law

Law: A physically realizable external configuration must possess a nonzero amplitude under its complete spacetime constraints; configurations that cannot consistently close upon themselves are excluded rather than retroactively altered.

Introduce a consistency operator C acting on the space of candidate external configurations:

$$C|E\blacksquare\blacksquare = |E\blacksquare\blacksquare \text{ for self-consistent configurations}$$

$$C|E\blacksquare\blacksquare = 0 \text{ for inconsistent configurations.}$$

The surviving state can then be represented schematically by

$$|\Psi_{allowed}\blacksquare = C|\Psi\blacksquare\ /\ \sqrt{\blacksquare}\Psi|C|\Psi\blacksquare$$

provided the denominator is nonzero. This is a toy projection/postselection picture. It should not be confused with a new experimentally established quantum law.

4. Why the Past Need Not Change

The distinction between *rewriting the past* and *restricting allowed amplitudes* is central. A literal retroactive rewrite would require a transformation such as

$$|\Psi(t_{past})\blacksquare \rightarrow |\Psi'(t_{past})\blacksquare$$

after the past had already been physically established. The proposed law instead says that a complete external configuration inconsistent with the global boundary conditions receives zero or vanishing amplitude. In that interpretation, there is no second version of the observed past waiting to be overwritten. The apparently alternative configuration was never selected as a complete self-consistent physical solution.

5. Relation to Closed Timelike Curves

The idea has a clear conceptual relationship to the Novikov self-consistency principle, under which local physical solutions in a spacetime containing closed timelike curves must be extendible to a globally self-consistent solution. Novikov and collaborators studied self-consistent solutions in time-machine spacetimes. ■cite■turn0search2■turn0search5■

Quantum formulations differ. Deutsch proposed a density-matrix fixed-point condition for quantum systems interacting with closed timelike lines, while later work explored postselection-based formulations. These approaches are mathematically and physically distinct, so the present law should be regarded as a synthesis-inspired toy framework rather than a claim that they are equivalent. ■cite■turn0search0■turn0search4■turn0search7■

6. A Machine-Level Interpretation

Stage	Hypothesized role
Electron-beam splitter	Creates spatial/beam alternatives that can later interfere.
Temporal magnetic synchronization	Defines a controlled temporal phase/interaction condition.
CTC/self-consistency region	Imposes the hypothetical global consistency constraint.
Synchrotron radiation	Provides a possible physical readout channel.
X-ray detector	Samples an observable consequence of the interaction.
Final collapse / selection	Represents the measurement of the surviving externally consistent outcome.

Under the proposed model, the machine does not need to send a conventional signal backward through time. Its defining operation would instead be a global consistency filter. A candidate external configuration is admitted only when its amplitude remains compatible with the machine's complete temporal boundary conditions.

7. A Global Amplitude Condition

Let F(E) represent the complete set of temporal, dynamical, and boundary constraints imposed on an external configuration E. A compact statement of the proposed law is

$$A(E) \neq 0 \Rightarrow F(E) = 0$$

In words: nonzero amplitude implies satisfaction of the global consistency condition. Conversely, if $F(E) \neq 0$, the idealized model assigns $A(E) = 0$. This formulation resembles a constraint or postselection rule. The physical mechanism capable of implementing such a rule is not supplied by the equation itself and remains an open question.

8. What the Law Would Predict

If the hypothetical machine existed and the proposed law applied, several qualitative predictions would follow. First, apparently paradoxical external outcomes would not require the observed past to change. Second, interference could depend on a global temporal consistency condition rather than only on local propagation. Third, an outcome that has a mathematically describable amplitude before imposing the global constraint could become unobservable after the constraint is applied. Finally, repeated measurements should reveal a reproducible statistical signature if the consistency operation were physically well defined.

9. Experimental Boundary

No known experiment establishes the existence of a macroscopic closed timelike curve, nor does the present paper establish that an electron beam, magnetic synchronization system, synchrotron source, or X-ray detector can create one. Existing work on closed-timelike-curve models includes theoretical and laboratory simulations, but these do not constitute construction of a macroscopic time machine. Accordingly, the proposed External-Space Self-Consistency Law should be treated as a speculative hypothesis and mathematical design principle.

10. Conclusion

The central idea is a shift in language from *changing histories* to *filtering external configurations*. In the proposed machine, amplitudes represent candidate external states, while a hypothetical global consistency condition determines which candidates can survive. The machine therefore need not 'repair' a past event after the fact. Instead, the complete spacetime description selects configurations that close consistently. The resulting picture is a machine governed not by retroactive rewriting, but by a proposed law of global amplitude compatibility.

References

1. I. D. Novikov et al., *Cauchy problem in spacetimes with closed timelike curves*, Physical Review D 42, 1915 (1990). ■cite■turn0search5■turn0search8■

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4. S. Lloyd et al., *Closed Timelike Curves via Postselection: Theory and Experimental Test of Consistency*, Physical Review Letters 106, 040403 (2011). ■cite■turn0search4■

Author's note: The External-Space Self-Consistency Law is introduced here as a speculative conceptual law for the proposed machine. It is not presented as an established law of physics.